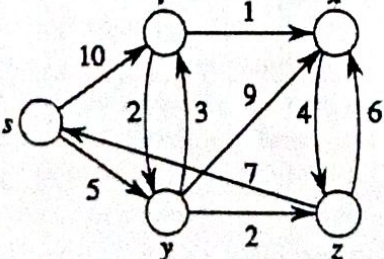


	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Information Technology Summer Term Examination Academic Year: 2024-25		
	Class: TY	Subject: Design and Analysis of Algorithms	Subject Code: PCC-IT308
	Date: 20.06.2025	Time: 10:00 to 13:00	Max. Marks: 100

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.
- Attempt any 5 questions

Q.N.	Questions	Marks	CO	BT
Q.1	a Explain merge sort algorithm in detail with example.	10	CO3	BT1
	b Explain insertion sort algorithm in detail with example.	10	CO3	BT1
Q.2	What is the recurrence relation for the following function "cal" <pre> int cal(int n) { if (n == 1) return 1; else return (cal(n - 1) + cal(n - 1)); } </pre> Evaluate the recurrence using method of substitution.	10	CO1	BT5
	Evaluate the following recurrence using method of substitution $T(n) = 2T(n-1) - 1$, when $n > 1$ $T(1) = 2$	10	CO1	BT5
Q.3	a Explain Master's method with its conditions Apply Master's method to following two recurrences for evaluation 1) $T(n) = 8T(n/2) + n^3$ 2) $T(n) = 16T(n/4) + n$	10	CO1	BT5
	b Explain binary search algorithm in detail with example	10	CO3	BT1
Q.4	a Explain quicksort algorithm in detail with example	10	CO3	BT1
	b Explain AVL tree with its rotations.	10	CO3	BT4
Q.5	a Explain Kruskal's algorithm for Minimum cost spanning tree with example	10	CO3	BT1
	b  Apply Dijkstra's Single Source shortest path algorithm to the given graph (Consider s as starting/source vertex)	10	CO3	BT3

Q.6	a	Using the hash function 'key mod 13', insert the following sequence of keys in the hash table 1, 28, 42, 53, 67,81,94,105,120,135. Use Linear Probing approach to handle collisions.	10	CO2	BT5
	b	Using the hash function 'key mod 13', insert the following sequence of keys in the hash table 1, 28, 42, 53, 67,81,94,105,120,135. Use separate chaining approach to handle collisions.	10	CO2	BT5
Q.7	a	Explain Backtracking approach of algorithms with an example	10	CO4	BT2
	b	Explain Sequential search and exhaustive search in brute force approach of algorithms.	10	CO4	BT2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Arrange the correctness of algorithm using inductive proofs and invariants.
CO 2	Understand the design of efficient algorithm and analyse the algorithms
CO 3	Describe, apply and analyse the complexity of certain divide and conquer, greedy and dynamic programming algorithm.
CO 4	Choose the proper suitable algorithmic design technique for solving any problem.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

Summer Term Examination
Academic Year: 2024-25

BTech (Information Technology)

Course: Python Programming

Third Year

Course Code: PCC-IT305

Date: 19-06-2025

Time: 10:00 to 13:00

Max. Marks: 100

Instructions: 1. Make suitable assumptions wherever necessary. 2. All Questions are compulsory.

Q.N.	Questions	Marks	CO IT3 05. X	BT X												
Q. 1	A Carefully analyze the following code snippets and predict the output? <u>Provide a valid justification for your answers. without it no marks will be awarded.</u> <table><tr><td>1</td><td><code>a = [1, 2, 3, 4] a.reverse() print(a[1])</code></td><td>2</td><td><code>a,*b,c = [4,5,6,7,8] print(a + len(b) + c)</code></td></tr><tr><td>3</td><td><code>def func(a,b=2,*args,c=3, **fun): print(a, b, args, c, fun) func(2, 0, 2, 5, d=19, e=6)</code></td><td>4</td><td><code>a = "summer" b = "su"+2*"m"+"er" c = "sum" + "mer" print(a is b, a is c)</code></td></tr><tr><td>5</td><td><code>class ST: x = [] e1 = ST() e2 = ST() e1.x.append(12) print(e2.x)</code></td><td>6</td><td><code>for n in range(1,7,2): if n == 4: break print("FUN:", n) else: print("FUN:", n)</code></td></tr></table>	1	<code>a = [1, 2, 3, 4] a.reverse() print(a[1])</code>	2	<code>a,*b,c = [4,5,6,7,8] print(a + len(b) + c)</code>	3	<code>def func(a,b=2,*args,c=3, **fun): print(a, b, args, c, fun) func(2, 0, 2, 5, d=19, e=6)</code>	4	<code>a = "summer" b = "su"+2*"m"+"er" c = "sum" + "mer" print(a is b, a is c)</code>	5	<code>class ST: x = [] e1 = ST() e2 = ST() e1.x.append(12) print(e2.x)</code>	6	<code>for n in range(1,7,2): if n == 4: break print("FUN:", n) else: print("FUN:", n)</code>	10	1-3	4, 5
1	<code>a = [1, 2, 3, 4] a.reverse() print(a[1])</code>	2	<code>a,*b,c = [4,5,6,7,8] print(a + len(b) + c)</code>													
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	B What will be the output of following python commands when executed on the python interpreter? <u>Provide a valid justification for your answers. without it no marks will be awarded.</u> <table><tr><td>1</td><td><code>>>> s={0, 12, 12, 13, 4, 4, 4} >>> s</code></td><td>2</td><td><code>>>>c = -4 - 3j; >>>abs(c)**2</code></td></tr><tr><td>3</td><td><code>>>> b = [1,2] * 3 >>> b</code></td><td>4</td><td><code>>>> a = 7, 8 >>> type(a)</code></td></tr><tr><td>5</td><td><code>>>>x = {1:2, 20:30, 3:4, 4:5,} >>>a = list(x.values())[:-1][0] >>>a + x[a-1]</code></td><td>6</td><td><code>>>>a = [2, 4, -3, 2] >>>sorted(a) >>>a[2]</code></td></tr></table>	1	<code>>>> s={0, 12, 12, 13, 4, 4, 4} >>> s</code>	2	<code>>>>c = -4 - 3j; >>>abs(c)**2</code>	3	<code>>>> b = [1,2] * 3 >>> b</code>	4	<code>>>> a = 7, 8 >>> type(a)</code>	5	<code>>>>x = {1:2, 20:30, 3:4, 4:5,} >>>a = list(x.values())[:-1][0] >>>a + x[a-1]</code>	6	<code>>>>a = [2, 4, -3, 2] >>>sorted(a) >>>a[2]</code>	10	1-2	5
1	<code>>>> s={0, 12, 12, 13, 4, 4, 4} >>> s</code>	2	<code>>>>c = -4 - 3j; >>>abs(c)**2</code>													
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Q. 2	A Explain the purpose of any five methods from the Python dict class. <i>For each method, provide a suitable example that clearly demonstrates how the method works and when it can be used.</i>	10	1	1-3												
	B Explain the purpose and working of the built-in int class constructors, i.e. int([x]) and int(x, base). <i>Specifically discuss each form with suitable examples. Also, mention the valid types of arguments accepted and any common errors that may occur.</i>	10	1	2-4, 6												
Q. 3	A Write a Python program to demonstrate exception handling for the cases such as division by zero, file not found, and invalid input (in case of user entering non-integer but integer value expected). Also explain how the try-except-finally block works in Python using your program.	10	2	3-4, 6												
	B Explain list comprehension, set comprehension, and dictionary comprehension in Python using suitable examples. <i>Your answer should clearly describe the syntax, purpose, and example with output for each comprehension.</i>	10	1	2, 6												
Q. 4	A Explain how inheritance works in Python with the help of an example involving classes Person, Faculty, and Student. Your example should: a) include a base class having some attributes and at least one method b) demonstrate how derived classes override the base class method	10	2-4	2, 6												

		c) show the use of public, protected, and private members in your classes Provide a complete Python program and provide its output.	10	2-4	2, 6
	B	Write a Python program (using either a GUI or command-line interface) to accept n numbers from the user and perform the following operations. a) Display the sum of all numbers. b) Display the maximum and minimum number. c) Display the list in ascending order. Provide a complete program with appropriate input/output. Also, explain the logic used for each operation.	10	2-4	2, 6
Q. 5	A	Develop a Python program (using either a GUI or command-line interface) that accepts a string input from the user and performs the following operations: a) Display the total number of vowels. b) Display the number of words and the longest word along with its length. c) Display whether the string is a palindrome or not. Provide a complete program with appropriate input/output. Also, explain the logic used for each operation.	10	2-4	2, 6
	B	Your friend is working on the following Python program to calculate the factorial of a number entered by the user. However, the program is <i>not working as expected</i> . It is producing errors during execution. <i>Your task is to identify as many errors as possible and provide corrected versions along with brief explanations for each identified mistake.</i> Carefully analyze your friend's solution provided below. # Your Friends Solution Starts here <pre>def Factorial(n): fact = 0 i = 1 while i <= n: fact *= i i++ return fact def displayresult(number, result): print("The factorial of " + number + "is" + result) n = input("Enter a number: ") if n < 0: print("Factorial is not defined") elif n == 0 or n == "0": print("Factorial of Zero is One") else: factorial_result = factorial(n) displayresult(n, factorial_result) # End of the code</pre>	10	3	4

"Python was created to be fun to use, and it has stayed that way throughout its evolution." - Guido Van Rossum
Creator of Python Programming language. All The BEST!

Course Outcomes (CO)

PCC-IT305.1	Describe the Numbers, Math functions, Strings, List, Tuples and Dictionaries in Python
PCC-IT305.2	Express different Decision-Making statements and Functions
PCC-IT305.3	Interpret Object oriented programming in Python
PCC-IT305.4	To design GUI Applications in Python and evaluate different database operations

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create